Homi Bhabha National Institute, Mumbai

(A University established under Section-3 of the UGC Act, 1956)

Programme conducted at

National Institute of Science Education and Research, Bhubaneswar

(An Off-Campus Centre of Homi Bhabha National Institute, Mumbai, India)

TRANSCRIPT OF RECORDS

Programme Name	: M.Sc. (5 Year Integrated)	**Discipline**	: Physics (Major) Computer Science (Minor)
Student Name	: SUMMIT BIKRAM NAYAK	**Enrollment No.**	: PHYS11202013047
Date of Joining	: 01/08/2020	**Date of Completion**	: 24/05/2025

Semester	Course Code	Course Name	Credits	Grade	Grade Point	SGPA
I	B101	BIOLOGY I: SCIENCE OF LIFE	6	BC	42	
	B141	BIOLOGY LABORATORY-I	3	BB	24	
	C101	CHEMISTRY I	6	CC	36	
	C141	CHEMISTRY LAB I	3	AB	27	
	CS141	PROGRAMMING AND DATA STRUCTURES LAB-I	3	BC	21	
	H101	INTRODUCTION TO ECONOMICS	4	AB	36	
	H109	TECHNICAL COMMUNICATION-I	4	AB	36	
	M101	MATHEMATICS I	6	BC	42	
	P101	MECHANICS AND THERMODYNAMICS	6	BC	42	
	P141	PHYSICS LABORATORY I	3	AA	30	
		Total	44		336	7.64
II	B102	BIOLOGY II: CELLULAR AND GENETIC BASIS OF LIFE	6	BC	42	
	B142	BIOLOGY LABORATORY-II	3	BC	21	
	C102	CHEMISTRY II	6	AB	54	
	C142	CHEMISTRY LAB II	3	AB	27	
	CS142	PROGRAMMING AND DATA STRUCTURES LAB -II	3	BB	24	
	H110	TECHNICAL COMMUNICATION-II	4	AB	36	
	H133	INTRODUCTION TO SOCIOLOGY	4	BB	32	
	M102	MATHEMATICS II	6	AB	54	
	P102	INTRODUCTORY PHYSICS II	6	AB	54	
	P142	PHYSICS LABORATORY II	3	BB	24	
		Total	44		368	8.36
III	H225	INTRODUCTION TO PSYCHOLOGY	4	AB	36	
	H235	SOCIOLOGY OF SCIENCE AND TECHNOLOGY	4	AA	40	
	M205	LINEAR ALGEBRA	8	BC	56	
	P201	CLASSICAL MECHANICS-I	8	BB	64	
	P202	MATHEMATICAL METHODS I	8	AB	72	
	P207	LINEAR OPTICS	4	AA	40	
	P241	GENERAL PHYSICS LAB	3	AA	30	
	P245	BASIC ELECTRONICS THEORY AND LAB	5	BB	40	
		Total	44		378	8.59
IV	CS202	DISCRETE STRUCTURES AND COMPUTATION	8	CD	40	
	H205	ENVIRONMENTAL ECONOMICS AND ENVIRONMENTAL IMPACT ASSESSMENT	4	BB	32	
	H227	ORGANIZATIONAL BEHAVIOUR	4	AB	36	
	P204	ELECTROMAGNETISM I	8	BC	56	
	P205	MATHEMATICAL METHODS II	8	BC	56	
	P206	QUANTUM MECHANICS I	8	BB	64	
	P246	ADVANCED ELECTRONICS THEORY AND LAB	4	BB	32	
		Total	44		316	7.18
V	CS301	DESIGN AND ANALYSIS OF ALGORITHMS	8	CD	40	
	P302	STATISTICAL MECHANICS	8	CD	40	
	P303	QUANTUM MECHANICS II	8	DD	32	
	P304	SPECIAL THEORY OF RELATIVITY	8	BB	64	
	P343	MODERN PHYSICS LABORATORY	3	BB	24	
	P345	OPTICS LABORATORY	3	AB	27	
	P346	COMPUTATIONAL PHYSICS LABORATORY	6	DD	24	
		Total	44		251	5.70
VI	CS460	MACHINE LEARNING	8	DD	32	
	P301	ELECTROMAGNETISM II	8	CC	48	
	P306	INTRODUCTION TO CONDENSED MATTER PHYSICS	8	CC	48	
	P307	NUCLEI AND PARTICLES	8	BC	56	
	P341	NUCLEAR PHYSICS AND INSTRUMENTATION LAB	6	BB	48	
	P347	SOLID STATE PHYSICS LABORATORY	6	BB	48	
		Total	44		280	6.36

Semester	Course Code	Course Name	Credits	Grade	Grade Point	SGPA
VII	P401	CLASSICAL MECHANICS II: MECHANICS OF CONTINUOUS MEDIA	8	CD	40	
	P441	INTEGRATED PHYSICS LABORATORY I	8	AB	72	
	P451	ADVANCED SOLID STATE PHYSICS	8	CD	40	
	P473	EXPERIMENTAL TECHNIQUES	8	DD	32	
		Total	32		184	5.75
VIII	CS457	PARAMETERIZED ALGORITHMS	8	DD	32	
	P443	INTEGRATED PHYSICS LAB II	8	BB	64	
	P452	COMPUTATIONAL PHYSICS	8	CD	40	
	P456	NONLINEAR OPTICS AND LASERS	8	DD	32	
	P457	GENERAL THEORY OF RELATIVITY AND COSMOLOGY	8	DD	32	
		Total	40		200	5.00
IX	CS201	THEORY OF COMPUTATION	8	CD	40	
	P405	ATOMS MOLECULES AND RADIATION	8	CD	40	
	P598	PHYSICS PROJECT	20	BC	140	
		Total	36		220	6.11
X	B306	EVOLUTIONARY BIOLOGY	8	CC	48	
	CS461	ADVANCED MACHINE LEARNING	8	CC	48	
	P475	SPECIAL TOPICS IN QUANTUM MECHANICS	8	CD	40	
	P599	PHYSICS DISSERTATION	20	BB	160	
		Total	44		296	6.73
			Total Credits = 416		CGPA : 6.80	

Date: 12/07/2025

Dean (AA)

Director

Checked by: APO/AO-I

Prepared by: UDC/OA MS

NISER Bhubaneswar

होमी भाभा राष्ट्रीय संस्थान
Homi Bhabha National Institute



होमी भाभा राष्ट्रीय संस्थान
एतद् द्वारा
विज्ञान स्नातकोत्तर
(५-वर्षीय एकीकृत कार्यक्रम)
की उपाधि
सम्मिट बिक्रम नायक
(नामांकन संख्या : PHYS11202013047)
(राष्ट्रीय विज्ञान शिक्षा एवं अनुसंधान संस्थान)
को विधिवत् रूप से इसके योग्य पाने तथा
६.८० सी जी पी ए प्राप्त करने पर
भौतिक शास्त्र
में
प्रदान करता है।

Homi Bhabha National Institute
hereby confers the degree of
Master of Science
(5-year Integrated Programme)
on
Summit Bikram Nayak
(Enrolment No. : PHYS11202013047)
(National Institute of Science Education and Research)
in
Physics
the student having been found duly qualified for the same with CGPA of 6.80

अध्यक्ष, विद्या परिषद् Chairman, Academic Council

अध्यक्ष, प्रबंधन परिषद् Chairman, Council of Management

Homi Bhabha National Institute is a Deemed to be University established under Section 3 of the UGC Act. 1956.
Regd. Office: Training School Complex, Anushaktinagar, Mumbai-400 094

Result approved on : June 19, 2025